

Sri Chaitanya IIT Academy.,India.

A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -1(Adv-2024-P1-Model)** **Date: 03-05-2026**

Time: 09.00Am to 12.00Pm **GTA-29** **Max. Marks: 180**

03-05-2026_Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT_Jee-Adv(2024-P1)_GTA-29_Syllabus,

PHYSICS : TOTAL SYLLABUS

CHEMISTRY : TOTAL SYLLABUS

MATHEMATICS : TOTAL SYLLABUS

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Name of the Student: _____ **H.T. NO:** | | | | | | | |

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JEE-ADVANCED-2024-P1-Model

Time:3Hr's **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 4)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 5 – 7)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 8 – 13)	Questions with **NON-NEGATIVE INTEGER** Type	+4	0	6	24
Sec – IV(Q.N : 14 – 17)	Matching Type	+3	-1	4	12
Total				**17**	**60**

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 18 – 21)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 22 – 24)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 25 – 30)	Questions with **NON-NEGATIVE INTEGER** Type	+4	0	6	24
Sec – IV(Q.N : 31 – 34)	Matching Type	+3	-1	4	12
				17	**60**

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 35 – 38)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 39 – 41)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 42 – 47)	Questions with **NON-NEGATIVE INTEGER** Type	+4	0	6	24
Sec – IV(Q.N : 48 – 51)	Matching Type	+3	-1	4	12
Total				**17**	**60**

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MATHEMATICS **Max Marks: 60**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

1. For a real parameter $a>0$, let $N(a)$ denote the number of solutions $x\in\left(0,\frac{\pi}{2}\right)$ satisfying $\sin^{-1}\left(\frac{2a\tan x}{1+a^2\tan^2 x}\right)=\frac{\pi}{2}-x$. Choose the correct statement about $N(a)$.

A) $N(a)=\begin{cases}1, & 0<a\le 2,\\ 2, & a>2\end{cases}$

B) $N(a)=\begin{cases}2, & 0<a<2,\\ 1, & a=2,\\ 2, & a>2\end{cases}$

C) $N(a)=\begin{cases}1, & 0<a<2,\\ 2, & a\ge 2\end{cases}$

D) $N(a)=\begin{cases}1, & 0<a<2,\\ 0, & a=2,\\ 2, & a>2\end{cases}$

2. Let Ω be the parabola $y^2=4x$. A point $P(h,k)$ moves on the circle $h^2+k^2=5$. If two distinct real tangents can be drawn from P to Ω, they touch Ω at A and B. Let R(P) be the bounded region enclosed by PA, PB and the parabolic arc $\overset{\frown}{AB}$. If S(P) denote the area of R(P), then the maximum possible values of S(P) is

A) $\frac{4}{3}5^{3/4}$ **B)** $\frac{1}{6}(2+4\sqrt{3})^{3/2}$ **C)** $\frac{9}{2}$ **D)** $\frac{19}{4}$

3. Three urns are given:

U_1 : 3 red balls and 1 blue ball,

U_2 : 2 red balls and 2 blue balls,

U_3 : 1 red ball and 3 blue balls.

An urn is selected at random. Then the following experiment is performed:

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i) Draw two balls successively without replacement and note their colours.

ii) If the two colours are same, keep both balls aside (do not return them), and then draw one more ball without replacement. (Total Draws =3)

iii) If the two colours are different, return both balls to the urn (restoring its original composition), and then draw three balls successively without replacement. (Total Draws =5)

Let E be the event that the total number of red balls drawn during the experiment is exactly 3, and the last ball drawn is red. Find $P(U_1|E)$.

(successively means continuing one after the other without any break)

A) $\frac{9}{13}$ **B)** $\frac{2}{3}$ **C)** $\frac{3}{5}$ **D)** $\frac{9}{11}$

4. For a real parameter a, define $f_a(x) = x^3 - 3x + a|x| (-1 \le x \le 2)$. Let $A(-1, f_a(-1))$ and $B(2, f_a(2))$. Let ℓ be the chord AB. Let N(a) be the number of points $c \in (-1,2)$ such that the tangent exists at $x = c$ for the curve $y = f_a(x)$ and is parallel to the chord ℓ.

Then, the set of all real a for which N(a) = 1 will be

A) $a \in \left(-\frac{27}{2}, \frac{9}{4}\right] \cup \left(0, \frac{9}{2}\right)$ **B)** $a \in \left(-\frac{27}{2}, -\frac{9}{4}\right] \cup \left[0, \frac{9}{2}\right)$

C) $a \in \left[-\frac{27}{2}, -\frac{9}{4}\right] \cup \left[0, \frac{9}{2}\right)$ **D)** $a \in \left(-\frac{27}{2}, -\frac{9}{4}\right) \cup \left[0, \frac{9}{2}\right)$

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.

For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

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5. For a real parameter $a \in (0,2\pi)$ and $n \in N$, define $z_n(a) = \left(1+\frac{ia}{n}\right)^n$,

$\theta_n(a) = Arg(z_n(a)) \in (-\pi,\pi]$ (principal argument). Define the sets

$S = \left\{a : (\theta_n(a))_{n\in N} \text{ is strictly increasing and } \lim_{n\to\infty} \theta_n(a) \in (0,\pi)\right\}$,

$K = \{a : \exists n \in N \text{ such that } z_n(a) \in R\}$.

Then which of the following is(are) true?

A) $S = (0,\pi)$

B) For every $a \in (0,2\pi)$, the sequence $(\theta_n(a))$ is strictly increasing

C) $K = \{n\tan(\pi/n) : n \in N, n \geq 3\} \cap (0,2\pi)$

D) For each fixed $a > 0$, the real sequence $u_n(a) = n\tan^{-1}\left(\frac{a}{n}\right)$ is strictly increasing in n and $\lim_{n\to\infty} u_n(a) = a$

6. The volume of the parallelepiped whose coterminous edges are represented by the vectors $2\vec{b}\times\vec{c}$, $3\vec{c}\times\vec{a}$ and $4\vec{a}\times\vec{b}$ where $\vec{a} = (1+\sin\theta)\hat{i} + \cos\theta\hat{j} + \sin 2\theta\hat{k}$,

$\vec{b} = \sin\left(\theta+\frac{2\pi}{3}\right)\hat{i} + \cos\left(\theta+\frac{2\pi}{3}\right)\hat{j} + \sin\left(2\theta+\frac{4\pi}{3}\right)\hat{k}$,

$\vec{c} = \sin\left(\theta-\frac{2\pi}{3}\right)\hat{i} + \cos\left(\theta-\frac{2\pi}{3}\right)\hat{j} + \sin\left(2\theta-\frac{4\pi}{3}\right)\hat{k}$ is 18 cubic units, then the values of θ, in the interval $\left(0,\frac{\pi}{2}\right)$, is/are

A) $\frac{\pi}{9}$ **B)** $\frac{2\pi}{9}$ **C)** $\frac{3\pi}{9}$ **D)** $\frac{4\pi}{9}$

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7. A hyperbola has center C and one focus at $P(10,24)$. If its two directrices are $5x+12y+26=0$ and $5x+12y-26=0$, then which of the following is/are incorrect?

A) $CP=26$

B) eccentricity $=\sqrt{13}$

C) $CP=24$

D) eccentricity $=\frac{\sqrt{13}}{2}$

SECTION-III
(NON-NEGATIVE INTEGER.)

This section contains **SIX (06)** questions.

- The answer to each question is a **NON-NEGATIVE INTEGER**.
- For each question, enter the correct integer corresponding to the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme

Full Marks : +4 If ONLY the correct integer is entered; Zero Marks : 0 In all other cases.

8. If the number of solution(s) of the equation $2\left[e^{x}\right]+4[lnx]+2[\sin x]=7$ is λ, then the value of $(2\lambda+3)$ is ____($[.]$ represents greatest integer function)

9. Twelve distinct students, numbered 1 to 12, are to be seated in a row of 12 seats numbered 1 to 12 from left to right (seat 1 is the leftmost). An arrangement is called admissible if all the following hold:

1) students 1, 2, 3, 4 appear from left to right in increasing order (i.e., pos(1) < pos(2) < pos(3) < pos(4), not necessarily consecutive)

2) Students 5 and 6 occupy adjacent seats.

3) There are exactly two students sitting between students 7 and 8.

If the number of admissible arrangements is γ, then the remainder when γ is divided by 1000 is ____ ((pos) means position)

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10. If $I=\begin{pmatrix}1&0&0\\0&1&0\\0&0&1\end{pmatrix}, P=\begin{pmatrix}4&3&4\\3&6&5\\4&5&9\end{pmatrix}, Q=\begin{pmatrix}a&b&c\\d&2&3\\e&4&5\end{pmatrix}$ for some nonzero real numbers a,b,c,d,e for which there is 3×3 matrix R with all entries being non zero real numbers such that $QR=RP$, where R is a non singular matrix. Then the value of $\det(Q-6I)$ is

11. The median of the following frequency distribution is ____

Class	0-10	10-20	20-30	30-40	40-50
Frequency	8	30	40	12	10

12. The value of the expression $\dfrac{\int_0^{2026}[x]dx}{\int_0^{2026}\{x\}dx}$, where $[x]$ and $\{x\}$ are integral and fractional parts of x is equal to

13. Let $T=\{0,1,2,.....,16\}$. For an ordered triple (x,y,z) of distinct elements of T, define $A(x,y,z)=\begin{pmatrix}1&1&1\\x&y&z\\x^2&y^2&z^2\end{pmatrix}$. Let N be the number of ordered triples (x,y,z) such that $\det(A(x,y,z))=240$. Find N.

SECTION – IV
(MATCHING TYPE)

This section contains **FOUR (04)** Matching List Sets.
- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists : **List-I** and **List-II.**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Five** entries (P), (Q), (R), (S) and (T).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks:+3 **ONLY** if the option corresponding to the correct combination is chosen;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: –1 In all other cases.

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14. Two players, P_1 and P_2, play a game against each other.

In every round of the game, each player rolls a fair die once, where the six faces of the die have six distinct numbers.

Let x and y denote the readings on the die rolled by P_1 and P_2, respectively

If $x > y$, then P_1 scores 5 points and P_2 scores 0 points

If $x = y$, then each player scores 2 points

If $x < y$, then P_1 scores 0 points and P_2 scores 4 points

Let X_i and Y_i be the total scores of P_1 and P_2, respectively, after playing the ith round

List – I		List – II	
I)	Probability of $(X_2 \geq Y_2)$ is	P)	$\frac{1}{216}$
II)	Probability of $(X_2 > Y_2)$ is	Q)	$\frac{11}{16}$
III)	Probability of $(X_3 = Y_3)$ is	R)	$\frac{95}{144}$
IV)	Probability of $(X_3 > Y_3)$ is	S)	$\frac{505}{864}$
		T)	$\frac{15}{32}$

The correct option is:

A) $I \to Q; II \to R; III \to T; IV \to S$ **B)** $I \to Q; II \to R; III \to P; IV \to S$

C) $I \to P; II \to R; III \to Q; IV \to S$ **D)** $I \to P; II \to R; III \to Q; IV \to T$

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15. Let matrix $A=\begin{bmatrix}\sqrt{3} & 1 & 0\\ 1 & -\sqrt{3} & 0\\ 0 & 0 & 2\end{bmatrix}$, then $\det\left(2A^T+adjA\right)$ is equal to a. If the volume of tetrahedron formed by the plane passing through $P(1,2,2)$ at a distance of 3 units from origin with coordinate planes is V, then the greatest integer less than or equal to V is b. Length of diameter of the director circle of a circle which touches both the tangents to the ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$ which cuts equal intercepts on coordinate axes is equal to c.

List – I		List – II	
P)	$2a+b$ is equal to	1)	0
Q)	$4a+b-c$ is equal to	2)	36
R)	$a+b-c$ is equal to	3)	20
S)	$6a+b-3c$ is equal to	4)	24
		5)	30

A) **P**-2; Q-4; R-4;S-1 **B)** **P**-5; Q-3; R-3; S-2

C) **P**-5; Q-3; R-3; S-1 **D)** **P**-4; Q-2; R-3; S-2

16. Match the following Column-I with Column-II

Column-I		Column-II	
I)	$\int\frac{dx}{\left(a^2-x^2\right)^2}=\frac{1}{2a^2}\int\frac{dx}{a^2-x^2}$	P)	$\frac{x}{2a^2\left(a^2-x^2\right)}+C$
II)	$\int\frac{dx}{x^2\sqrt{x^2-a^2}}=$	Q)	$\frac{\sqrt{x^2-a^2}}{a^2x}+C$
III)	$\int\frac{\sqrt{a^2-x^2}}{x^2}dx=$	R)	$-\frac{\sqrt{a^2-x^2}}{x}+\cos^{-1}\frac{x}{a}+C$
IV)	$\int\frac{\sqrt{x^2-a^2}}{x}dx=$	S)	$\sqrt{x^2-a^2}-a\sec^{-1}\frac{x}{a}+C$

A) $I\to Q;II\to R;III\to P;IV\to S$ **B)** $I\to P;II\to Q;III\to R;IV\to S$

C) $I\to P;II\to R;III\to Q;IV\to S$ **D)** $I\to P;II\to R;III\to S;IV\to Q$

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17. Let $f(x)$ is a quadratic polynomial satisfying functional equation $f(-1-x)=f(-1+x)$;

$\forall x \in R$ and $(f(1)-5)^2+(f(-1)-1)^2=f'(-1)$

(where [.] represent greatest integer function)

List – I		List – II	
P)	The value of $\left[\sin^{-1}\left(f(x)\right)\right]$ (whenever defined) is	1)	0
Q)	The value of $\left[1+\operatorname{sgn}\left(f(x)\right)\right]$	2)	1
R)	The value of $\left[\tan^{-1}\left(\frac{1}{f(x)}\right)\right]$ is	3)	2
S)	The value of $\left[\cot^{-1}\left(\frac{1}{2^{f(x)}}\right)\right]$ can be	4)	3
		5)	Not defined

A) $P\to 2;Q\to 3;R\to 1;S\to 2$ **B)** $P\to 2;Q\to 2;R\to 3;S\to 1$

C) $P\to 3;Q\to 2;R\to 2;S\to 3$ **D)** $P\to 1;Q\to 3;R\to 3;S\to 4$

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PHYSICS **Max Marks: 60**

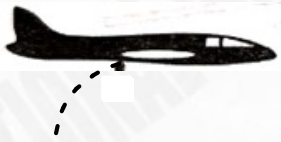
SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

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Negative Marks : −1 In all other cases

18. Figure shows a still photograph from a war movie. Bombs have been dropped from the plane B-52 at regular intervals. Airplane moves with constant speed in horizontal direction. Which figure may be true still photograph? [Assume that there is no wind.]

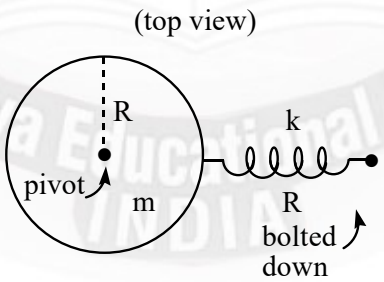
A) 

B) 

C) 

D) 

19. A uniform disk with mass m and radius R lies on a frictionless horizontal table and is free to rotate about a pivot at its center. A spring with spring constant k and relaxed length zero has one end attached to a point on the rim of the disk and the other end bolted down on the table at a distance R from the rim, as shown in figure. If the disk is initially in the equilibrium position shown and is then given a tiny angular kick, what is the angular frequency of small oscillations?



A) $\sqrt{\frac{2k}{m}}$ **B)** $2\sqrt{\frac{k}{m}}$ **C)** $\sqrt{\frac{k}{2m}}$ **D)** $\sqrt{\frac{k}{m}}$

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20. A thermally insulated container initially holds N_0 molecule of an ideal monatomic gas at temperature T_0. Molecules escape from the container through small holes in the wall, and it can be shown that in such a process at temperature T the average kinetic energy of the escaping molecules is 2kT. How many molecules remain in the container when their temperature has fallen to $\frac{1}{2}T_0$?

A) $\frac{N_0}{2}$ **B)** $\frac{N_0}{4}$ **C)** $\frac{N_0}{8}$ **D)** $\frac{N_0}{16}$

21. An ice cube of pure fresh water floats on pure fresh water in a glass. A huge iceberg, also of pure fresh water, floats on the salty water of ocean away from Antarctica. Neglecting the contribution due to the density of air, if the ice cube and the iceberg melt, then:

A) The water level rises both in the glass and in the ocean

B) The water level does not change in either case

C) The water level stays the same in the glass but rises in the ocean

D) The water level decrases in the glass but stays the same in the ocean

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SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
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Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.
For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

22. The particle displacement of a travelling longitudional wave is represented by $\xi = \xi(x,t)$. The midpoints of a compression zone and an adjacent rarefaction zone are represented by the letter 'C' and 'R'. Which of the following is true?

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A) $\left|\frac{\partial \xi}{\partial x}\right|_C = \left|\frac{\partial \xi}{\partial x}\right|_R$.

B) $\left|\frac{\partial \xi}{\partial t}\right|_C = \left|\frac{\partial \xi}{\partial t}\right|_R = 0$.

C) $(\text{Pressure})_C - (\text{Pressure})_R = 2\left|\frac{\partial \xi}{\partial x}\right|_C \times$ Bulk modulus of air.

D) Particles of air are stationary mid-way between 'C' and 'R'.

23. A disc of radius R with uniform positive charge density σ is placed on the $xy-$ plane with its centre at the origin. The coulomb potential along the z-axis is $V(z) = \frac{\sigma}{2\varepsilon_0}\left(\sqrt{R^2+z^2} - z\right)$. A particle of positive charge q is placed initially at rest at a point on the z-axis with $z = z_0$ and $z_0 > 0$. In addition to the coulomb force, the particle experiences a vertical $\vec{F} = -c\hat{k}$ with $c > 0$. Let $\beta = \frac{2c\varepsilon_0}{q\sigma}$. Which of the following statement(s) is(are) correct?

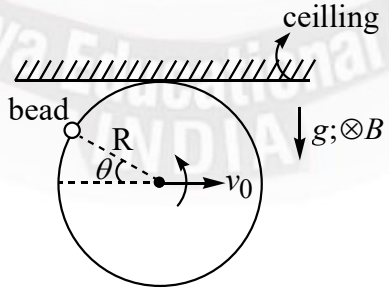
A) For $\beta = \frac{1}{4}$ and $z_0 = \frac{25}{7}R$, the particle reaches the origin.

B) For $\beta = \frac{1}{4}$ and $z_0 = \frac{3}{7}R$, the particle reaches the origin.

C) For $\beta = \frac{1}{4}$ and $z_0 = \frac{R}{\sqrt{3}}$, the particle returns back to $z = z_0$.

D) For $\beta > 1$ and $z_0 = 0$, the particle always reaches the origin.

24. An insulting ring of radius R, is rolled uniformly along the celling of a room with constant velocity v_0. A charged bead of mass m, charge q is placed over the ring. Entire system is moving in an uniform magnetic field as shown in the figure. The bead moves along with ring without any contact force from ring. $(\theta \neq 90^o)$



Choose the correct statements.

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A) The centripetal force on bead from the frame of reference of centre of mass of ring is qv_0B.

B) The magnetic field is $B=\frac{mg}{qv_0}$.

C) The radius of the ring is $R=\frac{mv_0}{qB}$.

D) The tangential acceleration of bead is $\frac{qv_0B\cos\theta}{m}$.

SECTION-III
(NON-NEGATIVE INTEGER.)

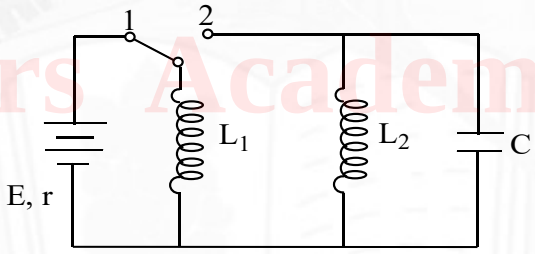
This section contains **SIX (06)** questions.

- The answer to each question is a **NON-NEGATIVE INTEGER**.
- For each question, enter the correct integer corresponding to the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme

Full Marks : +4 If ONLY the correct integer is entered; Zero Marks : 0 In all other cases..

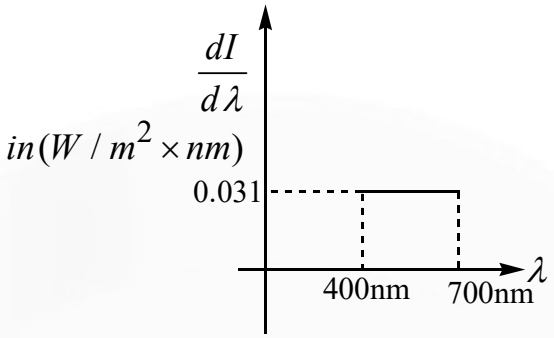
25. A circuit shown consists of two inductors of inductance L_1 and L_2, a capacitor of capacitance C, a battery of electromotive force E and internal resistance r and a switch. Initially the switch was in position 1 for a long time. Find the maximum current (in A) in the inductor L_2 after the switch is thrown to position 2.

(Given: E = 5V, $L_1=4mH$, $L_2=1mH$, $r=2\Omega$)

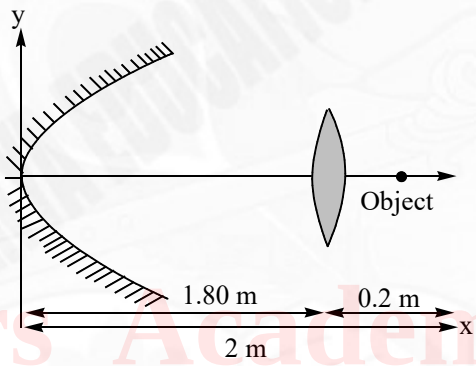


26. Scientists have made a light source whose spectral emissive power $\frac{dI}{d\lambda}$ is constant over visible range. Here I is intensity and λ is wavelength. In other words, $\frac{dI}{d\lambda}$ graph is shown below. This beam of area $1m^2$ is incident on emitter plate of photoelectric tube. The collector plate is sufficiently positive so that tube is in saturation mode. Assume that each capable photon liberates 1 electron. if work function is 2 eV, what is the current (in A) in the tube? Round off to nearest integer.

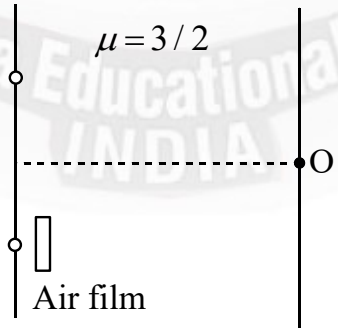
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27. A point object is kept at a distance of 2m from a reflecting surface $y^2 = 2x$. An equiconvex lens is kept at a distance of 1.80 m from the parabolic surface, The focal length of the lens is 20 cm. Position from origin of the image (in cm), after reflection from the surface is $10x$. Find the value of x.



28. YDSE is carried out in a liquid of refractive index $\mu = 3/2$. A thin film of air is formed in front of the lowest slit as shown in the figure. A maxima is formed at point O which is mid point of screen. If thickness of air film is $4\mu m$ then find order of maxima at point O. Wavelength of light in air is $4000\,\mathring{A}$.



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29. The electric field in a plane electromagnetic wave is given by $\vec{E} = 200\cos\left[\left(\frac{0.5\times10^3}{m}\right)x - \left(1.5\times10^{11}\frac{rad}{s}\times t\right)\right]\frac{V}{m}\hat{j}$. IF the wave falls normally on a perfectly reflecting surface having area of $100\,cm^2$. If the radiation pressure exerted by a E.M. wave on the surface during a 10minute exposure is $\frac{x}{10^9}\frac{N}{m^2}$. Find the value of x.

30. A stable star of radius R has a volume mass density profile $\rho(r) = \alpha(1 - r/R)$. Here, "stable" means that the star doesn't collapse under its own gravity. If the internal pressure at the core is provided solely by the radiation of photons and is equal to $\frac{4\sigma}{3c}T_c^{\,4}$. Assume the star is a perfect black body and treat photons as a classical ideal gas. Given α is a positive constant, σ is stephen's constant, c is speed of light, G is gravitational constant and T_c is the temperature at the core. If $T_c = \left[\frac{5}{x}\frac{\pi G\alpha^2 R^2 c}{\sigma}\right]^{1/4}$. Find x?

SECTION – IV
(MATCHING TYPE)

This section contains **FOUR (04)** Matching List Sets.
- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists : **List-I** and **List-II.**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Five** entries (P), (Q), (R), (S) and (T).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks:+3 **ONLY** if the option corresponding to the correct combination is chosen;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: –1 In all other cases.

31. Column – I describes some situations in which a small object moves, column – II describes some characteristics of these motions. Match the situations in Column – I with the characteristics in Column – II.

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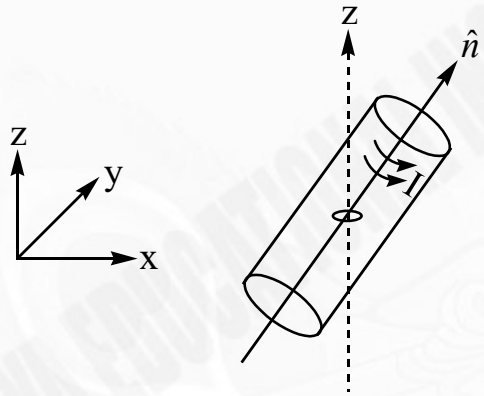
Column – I		Column – II	
A)	The object moves on the x-axis under a conservative force in such a way that its "speed" and "position" satisfy $\upsilon = c_1\sqrt{c_2 - x^2}$, where c_1 and c_2 are positive constant.	P)	The object executes a simple harmonic motion.
B)	The object moves on the x-axis in such a way that its velocity and its displacement from the origin satisfy $\upsilon = -kx$, where k is a positive constant.	Q)	The object does not change its direction.
C)	The object is attached to one end of a mass-less spring of a given spring constant. The other end of the spring is attached to the ceiling of an elevator initially everything is at rest. The elevator starts going upwards with a constant acceleration a. The motion of the object is observed from the elevator during the period it maintains this acceleration.	R)	The kinetic energy of the object keeps on decreasing.
D)	The object is projected from the earth's surface vertically upwards with a speed $2\sqrt{\frac{GM_e}{R_e}}$, where M_e is the mass of the earth and R_e is the radius of the earth. Neglect forces from objects other than the earth.	S)	The object can change its direction only once.

A) $A \to P; B \to QR; C \to P; D \to QR$ **B)** $A \to P; B \to Q; C \to PQ; D \to QS$

C) $A \to P; B \to QR; C \to S; D \to PQ$ **D)** $A \to Q; B \to Q; C \to P; D \to QR$

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32. A small circular loop of area A and resistance R is fixed on a horizontal $xy-$ plane with the center of the loop always on the axis $\hat{n}$ of a long solenoid. The solenoid has m turns per unit length and carries current I counterclockwise as shown in the figure. The magnetic field due to the solenoid is in $\hat{n}$ direction. List – I gives time depends of $\hat{n}$ in terms of a constant angular frequency ω. List – II gives the torques experienced by the circular loop at time $t=\frac{\pi}{6\omega}$, Let $\alpha=\frac{A^2\mu_0^2m^2I^2\omega}{2R}$. Which one of the following options is correct?



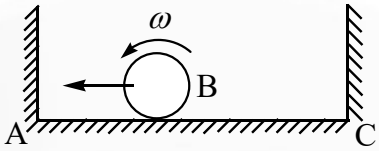
List – I		List – II	
A)	$\frac{(\sin\omega t\hat{j}+\cos\omega t\hat{k})}{\sqrt{2}}$	P)	0
B)	$\frac{(\sin\omega t\hat{i}+\cos\omega t\hat{j})}{\sqrt{2}}$	Q)	$-\frac{\alpha}{4}\hat{i}$
C)	$\frac{(\sin\omega t\hat{i}+\cos\omega t\hat{k})}{\sqrt{2}}$	R)	$\frac{3\alpha}{4}\hat{i}$
D)	$\frac{(\cos\omega t\hat{i}+\sin\omega t\hat{k})}{\sqrt{2}}$	S)	$\frac{\alpha}{4}\hat{j}$
		T)	$-\frac{3\alpha}{4}\hat{j}$

A) $A\to Q;B\to P;C\to S;D\to T$ **B)** $A\to S;B\to T;C\to Q;D\to P$

C) $A\to Q;B\to P;C\to S;D\to R$ **D)** $A\to T;B\to Q;C\to P;D\to R$

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33. A ball of mass m and radius R is rolled towards a wall A as shown. Wall A is smooth but wall C is rough. AB is rough but BC is smooth, ABC is horizontal ground. Consider the following during first cycle of motion only. All collisions are elastic.



Column – I		Column – II	
A)	During motion from A to B	P)	The friction may oppose its rotation
B)	During motion from C to B	Q)	Angular momentum about B is constant
C)	During collision with wall A	R)	Ball may start rotating clockwise
D)	During collision with wall C	S)	Mechanical energy may decrease

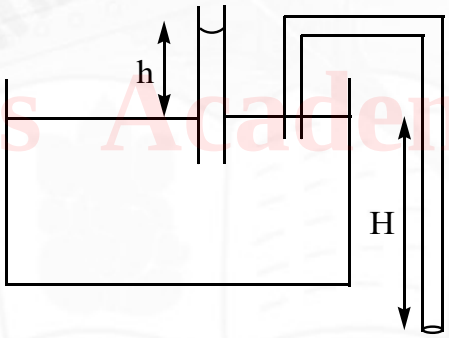
A) $A \to PQRS; B \to Q; C \to PQR; D \to RS$

B) $A \to PQRS; B \to Q; C \to PRS; D \to RS$

C) $A \to PQ; B \to P; C \to QR; D \to RS$

D) $A \to PQ; B \to R; C \to QRS; D \to R$

34. Two capillary tubes of the same cross-section are lowered into a vessel with water as shown in figure. The water in straight tube rises to a height h. The lower end of the bent tube is at a depth H below the level of the water in the vessel. What will be the shape of meniscus at the lower end of the bent tube (as seen from below) in different cases?



List – I		List – II	
P)	H = h	1)	Convex
Q)	0 < H < h	2)	Concave
R)	H = 0	3)	Flat
S)	H < 0 (the end of the bent tube is above the water level)	4)	Water flow out

A) $P \to 2; Q \to 2; R \to 3; S \to 4$ **B)** $P \to 3; Q \to 2; R \to 4; S \to 1$

C) $P \to 1; Q \to 4; R \to 3; S \to 2$ **D)** $P \to 1; Q \to 1; R \to 3; S \to 2$

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CHEMISTRY **Max Marks: 60**

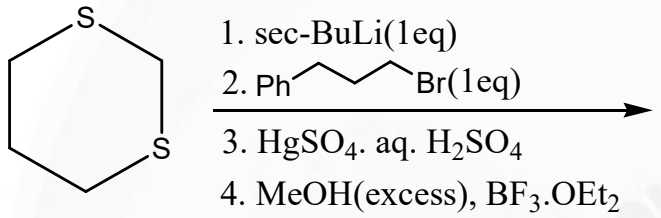
SECTION – II
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

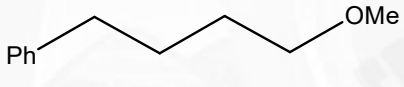
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

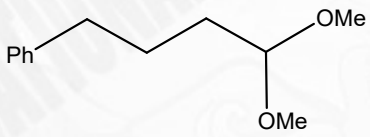
Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

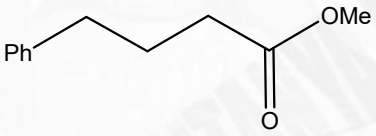
35. The major product formed in the following reaction

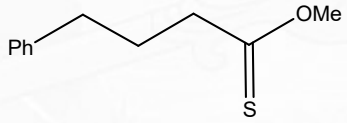


is

A) 

B) 

C) 

D) 

36. Two liquids A and B are mixed at temperature T in a certain ratio to form an ideal solution. It is found that the partial pressure of A, i.e., P_A is equal to P_B, the pressure of B for liquid mixture. What is the total pressure of the liquid mixture in terms of P_A^0 and P_B^0 ?

A) $\frac{P_A^0 P_B^0}{P_A^0 + P_B^0}$ **B)** $\frac{2P_A^0 P_B^0}{P_A^0 + P_B^0}$ **C)** $\frac{2P_A^0}{P_A^0 + P_B^0}$ **D)** $\frac{2P_B^0}{P_A^0 + P_B^0}$

37. You are given the following cell at 298K

$$Zn\left|Zn^{2+}(aq.)\right\|HCl_{(aq.)}\left\|H_{2(gas)}\right|Pt \text{ with } E_{cell} = 0.701V$$

$0.01M$ $1.0lit$ $1.0atm$

$E^0_{Zn^{2+}/Zn} = -0.76V$. Which of the following amounts of $NaOH$ (equivalent weight=40) will just make the pH of cathodic compartment to be equal to 7.0:

A) 0.4grams **B)** 4 grams **C)** 10 grams **D)** 2 grams

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38. Which of the following products are obtained, when $Na[HXeO_4]$ is allowed to stand in aq NaOH?

I) Xe **II)** $Na_4[XeO_6]$ **III)** H_2 **IV)** O_2

A) II and IV only **B)** II, III and IV only

C) II and III only **D)** I, II and IV only

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

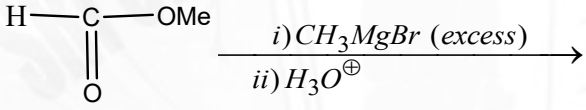
Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.

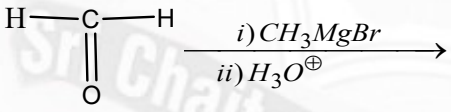
For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

39. Reactions in which product can show positive Haloform test is(are):

A) $PhCN \xrightarrow[ii)H_3O^{\oplus}]{i)CH_3MgBr\ (excess)}$

B) $Ph-COOEt \xrightarrow[ii)H_3O^{\oplus}]{i)\ CH_3MgBr\ (excess)}$

C) 

D) 

40. The correct option(s) related to adsorption processes is (are)

A) Chemisorption results in a multimolecular layer.

B) The enthalpy change during physisorption is in the range of 20 to $40\ kJ\ mol^{-1}$.

C) Chemisorption is an exothermic process.

D) Lowering the temperature favors physisorption processes.

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41. In which of the following, 6-membered ring formation takes place during hydrogen-bond formation?

A) $\left[Ni(dmg)_2\right]$ **B)** Chloral hydrate

C) Salicylaldehyde **D)** ortho-nitrophenol

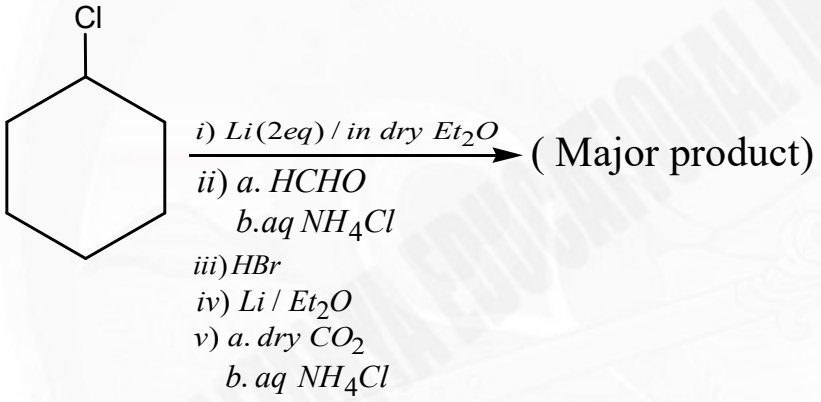
SECTION-III
(NON-NEGATIVE INTEGER.)

This section contains **SIX (06)** questions.

- The answer to each question is a **NON-NEGATIVE INTEGER.**
- For each question, enter the correct integer corresponding to the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme

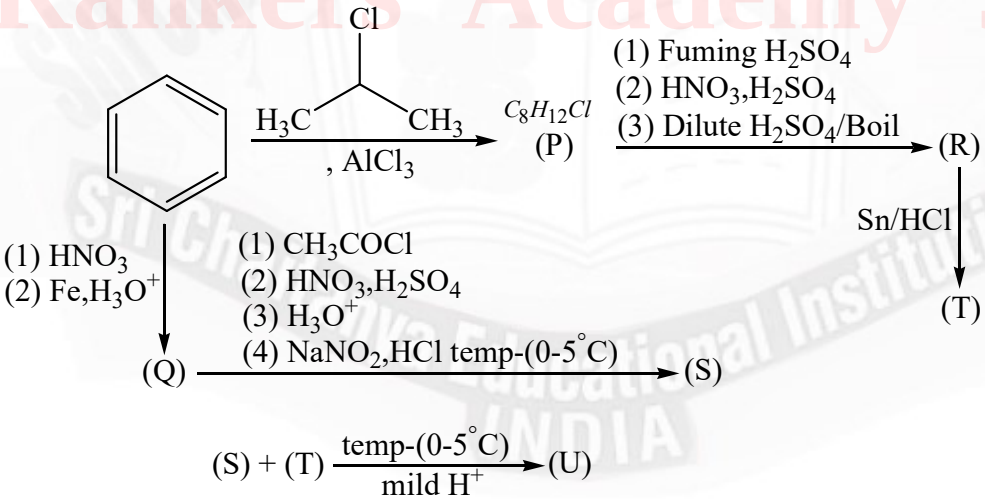
Full Marks : +4 If ONLY the correct integer is entered; Zero Marks : 0 In all other cases.

42.



How many different hydrogens are present in the major product?

43.



$$(S) + (T) \xrightarrow[\text{mild } H^+]{\text{temp-}(0\text{-}5^\circ C)} (U)$$

If x is the total number of nitrogen atoms present in compound(U) and y is the number of different hydrogens present in the compound (T). Then find the value of x+y?

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44. 3.0 moles of an ideal monatomic gas undergoes two processes:
Isothermal and reversible expansion from 4L to 8L at 400K
Then isochoric cooling until the pressure becomes 1/8 of the initial pressure
Calculate the magnitude of total entropy change (ΔS) of the system in J/K. Give the value of nearest integer
Given: $R = 8.314\, J/mol.K$, $\ln 2 = 0.7$

45. At 300K, the stopping potential for the photoelectrons being emitted from a metal surface of work function 5.4 eV is equal to the standard electrode potential of the cell;
$Ag(s)\left|AgBr(s)\right|KBr(aq)\left\|KCl(aq)\right|AgCl(s)\left|Ag(s)\right.$
When the wavelength of the incident photons on the same metal plate is halved, the wavelength of the most energetic photoelectrons emitted is $x\,\overset{0}{A}$. Find x (rounded off to the nearest integer)
Given $K_{SP}(AgCl) = 10^{-10}$, $K_{SP}(AgBr) = 10^{-15}$ *and* $\dfrac{2.303RT}{F} = 0.06$

46. Consider the following:
$HClO_4$, F_2, Cl_2O, Br_2, Cl_2O_7 and Cl_2.
If 'x' is the number of compounds which undergo disproportionation with NaOH, then the value of 'x' is____.

47. 3rd element of 12th vertical column and 3rd element of 17th vertical column are liquids with atomic numbers Z_1 and Z_2, but 3rd element of 13th vertical column is summer liquid with atomic number Z_3. Then the value of $(Z_1 + Z_2 + Z_3)$ is_____. (consider periodic table)

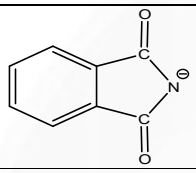
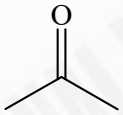
SECTION – IV
(MATCHING TYPE)

This section contains **FOUR (04)** Matching List Sets.
- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists : **List-I** and **List-II.**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Five** entries (P), (Q), (R), (S) and (T).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks:+3 **ONLY** if the option corresponding to the correct combination is chosen;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: –1 In all other cases..

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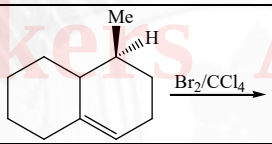
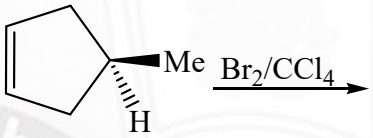
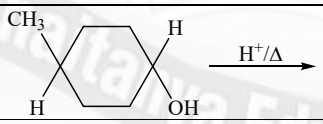
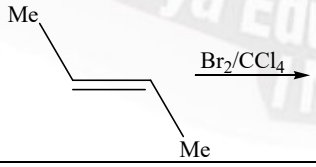
48. For the reaction using suitable reactant and suitable nucleophile what will be the correct match

List-I		List-II	
P)	MeS^-	1)	Reacts with CH_3CH_2Cl to give SN^2 product
Q)	$NC^- + NH_4Cl$	2)	Reacts with CH_3CHO followed by hydrolysis to give alanine.
R)		3)	Reacts with CH_3-CH_2-Cl followed by NH_2NH_2 gives $EtNH_2$
S)	$n-BuLi$	4)	Reacts with  followed by acetone to give hexan-2-ol.
		5)	Reacts with chloro benzene followed by H_3O^+ give aniline.

A) $P \to 1; Q \to 2; R \to 3; S \to 4$ **B)** $P \to 5; Q \to 2; R \to 1; S \to 4$

C) $P \to 5; Q \to 4; R \to 3; S \to 1$ **D)** $P \to 1; Q \to 4; R \to 5; S \to 4$

49. Match List.

List – I		List – II (About product mixture)	
P)		1)	Optically active
Q)		2)	Diastereoisomers
R)		3)	Optically inactive
S)		4)	Meso

A) $P \to 1,2; Q \to 3; R \to 3; S \to 3,4$ **B)** $P \to 1,3; Q \to 2; R \to 3; S \to 3,4$

C) $P \to 1,2; Q \to 3; R \to 1; S \to 2,3$ **D)** $P \to 1,2; Q \to 1; R \to 3; S \to 3,4$

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50. Match the following

Column-I		Column-II	
A)	$N_2O_4(g) \rightleftharpoons 2NO_2(g)$ On increases in volume at constant temparature	**p)**	No equilibrium shift
B)	$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$ On increases in volume at constant temparature	**q)**	Density of reacting mixture decreases
C)	$2NO(g) + Br_2(g) \rightleftharpoons 2NOBr(g)$ On decrease in pressure at constant temperature	**r)**	Average molar mass of reacting mixture increases
D)	$2ICl_3(g) \rightleftharpoons I_2(g) + 3Cl_2$ On increase in pressure at constant temperature	**s)**	Average molar mass of reacting mixture decreases
		t)	Reaction moves in forward direction

A) $A-qst;B-p;C-qs;D-r$ **B)** $A-qst;B-p;C-r;D-qs$

C) $A-qst;B-p;C-st;D-qs$ **D)** $A-qst;B-pq;C-st;D-r$

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51. Match the species given in Column – I with the character given in Column – II.

Column – I		Column – II	
P)	Hydrogen hypo phosphate ion $\left(H_2P_2O_6^{2-}\right)$ (Ignore isohypophosphoric acid structure)	1)	All hydrogens are ionizable
Q)	Hydrogen pyrophosphate ion $\left(H_2P_2O_7^{2-}\right)$	2)	Number of acidic hydrogens (replacable) present is 2
R)	Pyrosulphurous acid $\left(H_2S_2O_5\right)$	3)	sp^3 hybridised central atom(Phosphorus or Sulphur)
S)	Hydrogen peroxy mono sulphate ion $\left(HSO_5^-\right)$	4)	Overall non-planar structure
		5)	Can act as oxidizing agent

A) $P-1234, Q-345, R-1234, S-12345$

B) $P-124, Q-2345, R-134, S-1345$

C) $P-124, Q-34, R-134, S-12345$

D) $P-1234, Q-1234, R-1234, S-1345$

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